Kangrui Zhou

zhoukangruinudt@163.com — Personal Homepage

EDUCATION

National University of Defense Technology, Changsha, China — Sep 2023 — Dec 2025
Master in Aerospace Science and Technology — Cumulative GPA: 3.54/4.00
Selected Courses: *Principles and Practice of Artificial Intelligence (A), Meteorological and Oceanographic Information Processing Experiment (A), Computational Fluid Dynamics (A)*
Thesis Title: Deep Symbolic Regression-Driven Discovery and Mapping Learning of Thermo-Mechanical Coupling Models for Aerospace Vehicles

National University of Defense Technology, Changsha, China — Sep 2019 — July 2023
Bachelor in Aerospace Engineering — Cumulative GPA: 3.63/4.00
Selected Courses: *Advanced Mathematics (A), College Physics (A), Aerodynamics (A), Methods of Mathematical Physics (A)*

EXPERIENCE

Huawei Technologies Co., Ltd., Nanjing, China — Feb 2026 — Oct 2026
AI Software Engineer
Developed AI Agent applications for the data-communication vertical domain and Java back-end services.

PUBLICATIONS

Rapid Prediction of Thermal Stress via Domain Decomposition-based Hybrid Fourier Neural Operator
Kangrui Zhou, Wei Peng, Xiaoya Zhang, Xu Liu, Wen Yao
Engineering Applications of Artificial Intelligence, IF:8.0, 2025. (Published)

Event-based Depth Estimation with Dense Occlusion
Kangrui Zhou, Taihang Lei, Banglei Guan, and Qifeng Yu
Optics Letters, IF:3.1, 2024. (Published)

PROJECTS

Discovery and Mapping Learning of High-Efficiency Thermo-Mechanical Predictive Models for Cross-Domain Variable-Configuration Vehicles — Jul 2024 — Dec 2025

- Developed a two-stage deep symbolic regression method with a search space that decouples symbols from parameters: the first stage generates expression skeletons by combining mathematical operators with physical quantities, and the second dynamically embeds parameters into the expression tree. Reduced MAE by 50% on a two-dimensional rarefied flow problem.
- Designed a hybrid Fourier neural operator that employs Monte Carlo tree search-based domain decomposition to split the field into high- and low-frequency sub-regions, learns their mappings separately, and enforces interface continuity through a dedicated loss function. Reduced MAE by 27.8% on the heating prediction of three satellite motherboards.

SELECTED AWARDS AND HONORS

Outstanding Student (Top 10%), National University of Defense Technology — Jul 2020

Second Prize Scholarship of Academic Excellence, National University of Defense Technology — Jan 2025

National First Prize of China College Students' Advanced Graphics Technology and Product Information Modeling Innovation Competition — Aug 2022

National Second Prize of China Graduate Future Flight Vehiele Innovation Competition — Nov 2024

National Third Prize of China College Students' Energy Conservation and Emission Reduction Social Practice and Science and Technology Competition — Aug 2022

SKILLS

- **Programming & Software:** Python, Java, SolidWorks, Pointwise, Codex